

LYCANS AERODESIGN TEAM

Training Task 03

Avionics Software Engineering & Safety Systems Implementation

Deadline: Aug 24, 2026 – 11:59 PM
Submission: GitHub Repository

1 Objective

This task evaluates your practical software implementation skills for embedded flight control systems and ground station software.

You are required to:

- Implement all 5 programming modules covering C flight control logic and Python OOP architectures.
- Organize your implementation using the strict multi-language directory layout.
- Use GitHub Desktop to manage your assignment branch and version control.
- Submit your completed source files by opening a Pull Request on GitHub.

2 Required Tasks

You must complete all 5 engineering modules assigned during the workshop session:

1. **Task 1: Flight Controller Altitude Hold Logic (C)** – Implement core safety-critical logic in C evaluating arming status, low battery thresholds, and altitude deadbands.
2. **Task 2: Pre-Flight Readiness Engine (Python)** – Develop a flight parameter validation engine parsing aircraft fleets and reporting diagnostics.
3. **Task 3: Aircraft Object-Oriented Class (Python)** – Construct a robust Python class encapsulating telemetry state tracking and position history.
4. **Task 4: Sensor Package Modularization & File I/O (Python)** – Build a modular parsing engine reading raw sensor text logs and saving formatted security alerts.
5. **Task 5: Aircraft Subclasses & Inheritance (Python)** – Extend base functionality to support specialized Fixed-Wing and VTOL aircraft models using class inheritance.

3 File Structure Requirements

All code files must be placed inside your training repository under the `task-03` folder, strictly adhering to the following structure:

```
lycans_aircraft/
```

```
python/
```

```
    main.py  
    aircraft.py  
    sensors.py  
    safety.py
```

```
    data/  
        telemetry.txt  
        flight.log
```

```
c/  
    flight_controller.c
```

Ensure all filenames and directory paths match this specification exactly.

4 Submission Instructions

Submit your work using **GitHub Desktop**:

4.1 1. Create Branch in GitHub Desktop

1. Open GitHub Desktop and select your assignment repository. Use the same personal training repository you set up previously. Do not create a new one.
2. Click **Current Branch** at the top bar and select **New Branch**.
3. Name your branch:

```
task-03-avionics-software
```

4. Create the branch based on `main` and click **Publish branch**.

4.2 2. Implement Your Code

Add all files and directories locally according to the required file structure. Ensure:

- `flight_controller.c` compiles without warnings or errors.
- Python scripts run cleanly with expected output files.

4.3 3. Commit and Push

1. Open GitHub Desktop; you will see all new/modified files listed on the left panel.
2. Ensure all created files are checked.
3. Type a descriptive commit message in the summary box (e.g., `Implement Task 03 C and Python modules`).
4. Click **Commit to task-03-avionics-software**.
5. Click **Push origin** at the top bar to upload your changes to GitHub.

4.4 4. Open Pull Request

1. In GitHub Desktop, click the ****Create Pull Request**** button (or press Ctrl + R / Cmd + R).
2. This will open your web browser on GitHub.
3. Ensure base branch is set to `main` and compare branch is `task-03-avionics-software`.
4. Title your Pull Request: `Task 03 -- Avionics Software Engineering`.
5. Click ****Create Pull Request****.

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Do not merge your Pull Request. The evaluation team will review your submission directly on GitHub.

Build clean code. Test thoroughly. Submit on time.